

The selling-oriented versus customer-oriented retail chatbot and effects on customer satisfaction

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ARTICLE INFO

Keywords:

Retail chatbots
Selling orientation
Customer orientation
Warmth
Competence
Attribution of theory of mind
Customer satisfaction

ABSTRACT

Chatbots in e-retailing settings have so far mainly been examined in their role as service providers, although there is a growing pressure on human service employees to engage in selling activities. The current study addresses this gap by examining an online retail chatbot with a selling orientation. Contrasted with a customer-oriented approach in two between-subjects experiments, the study shows that customer satisfaction was significantly reduced when the chatbot used a selling orientation. This effect was mediated by perceived chatbot warmth and competence as well as attributions of theory of mind to the chatbot. When agent identity (chatbot vs. human employee) was manipulated in the second experiment, the results remained invariant.

1. Introduction

As online sales increase for retailers, many of them have introduced various forms of chatbots (i.e., digital agents capable of conducting a conversation in natural language) on their websites (Kim and Hur, 2024; Lee and Li, 2023; Zhang et al., 2025). This, then, is a manifestation of how artificial intelligence increasingly influences service encounters, which raises fundamental questions about the future of services (Morati and Viana Mundstock Freitas, 2026). Chatbots have been subject to several studies, and it is beginning to become clear what these bots should do and not do in their interactions with customers in order to become positively evaluated and to create customer satisfaction. The chatbot's conversational cues (Xu et al., 2023), conversational style (Bhattacharjee et al., 2025), interactivity (Lee and Li, 2023), gender, name and job title (Lu and Zhang, 2025), and emotional expressions (Zhang et al., 2025) are examples of chatbot attributes that have been assessed in empirical studies.

So far, however, the focus in existing studies has been on the service that chatbots offer. For instance, it has been argued that chatbots can deliver predictable and homogenous service interactions with minimal errors (Sands et al., 2021) and that they can handle large volumes of customer inquiries (Lu and Zhang, 2025). They can also enhance the service experience (Kull et al., 2021) and improve service efficiency (Xu et al., 2023; Zhang et al., 2025). Specific examples of services that chatbots can provide are to answer questions, solve problems (Bock et al., 2020), and provide detailed descriptions of products (Kim and

Hur, 2024).

Yet, in human-to-human service encounters, there is a growing pressure on employees to engage in explicit selling activities. In addition, salespersons are expected to provide service (de Ruyter et al., 2020). This creates a challenging situation, because serving customers and selling to them are regarded as independent, siloed activities in many firms. And forcing service employees to pursue selling can have a detrimental impact on their motivation (ibid.). Chatbots may be a solution to integrating selling and service, so it is expected that firms will attempt to increase chatbots' selling activities. To date, however, it is unclear from existing studies how customers would react when faced with a chatbot that is actively engaged in selling.

The current study addresses this gap by examining effects on customers when chatbots take on the task of selling. Our theoretical point of departure is the notion of a selling orientation, typically conceptualized in relation to a customer orientation. Since the distinction between the two orientations was launched by Saxe and Weitz (1982), several studies have examined them with respect to salespersons in human-to-human contexts (e.g., Anaza et al., 2023; Brown et al., 1991; Hansen et al., 2016; Itani et al., 2022; Kadic-Magljajic et al., 2017; Locander et al., 2023; Thomas et al., 2001). To date, however, they have not been examined when the firm representative is a non-human agent (such as a chatbot).

Existing evidence – from a human-to-human setting – indicates that the salesperson's selling orientation is inferior to a customer-oriented approach. More specifically, a selling orientation reduces customer

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<https://doi.org/10.1016/j.jretconser.2026.104805>

Received 11 November 2025; Received in revised form 12 March 2026; Accepted 13 March 2026

Available online 18 March 2026

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satisfaction with the salesperson (Goff et al., 1997), increases conflicts with the buyer, and enhances buyer propensity to leave (Anaza et al., 2023). It also attenuates repurchase intentions (Kadic-Magljajlic et al., 2017). In contrast, a customer orientation boosts satisfaction with the salesperson (Brady and Cronin Jr, 2001; Goff et al., 1997; Homburg et al., 2011), sales performance (Kuo et al., 2023), and value creation (Guenzi et al., 2011). More generally, in the literature review by Zhao (2022), it is argued that the positive effects of a customer orientation are largely undisputed.

However, quite some time has passed since several of these studies were carried out. Given that many firms have adopted a customer-oriented approach, it is possible that this approach is seen by contemporary customers as a coercion tactic with the ultimate goal of selling. Anaza et al. (2023), for example, did not find that the salesperson's customer orientation reduced the buyer's propensity to leave. It is also possible that a selling orientation is perceived as more efficient and less privacy-invasive, in the sense that it may not involve an extensive set of questions about personal matters asked by the agent who is making attempts to identify the customer's needs. Indeed, Guenzi et al. (2011) argue that the widespread bad reputation of a selling orientation may not be justified. Moreover, given that it can be fun and entertaining to encounter a hard-selling huckster (Brown, 2001), it may be even more so in the relatively novel situation in which the agent is non-human. After all, we humans have a long history of being fascinated by (humanlike) non-humans, and to sell is indeed a very human activity.

In the light of this, the purpose of the current study is to examine what happens when chatbots in a retail setting engage in selling activities within the frame of an online service encounter. Our dependent variable is customer satisfaction with the chatbot. The study also unpacks mediating mechanisms in terms of perceived chatbot warmth and competence and attributions of theory of mind to the chatbot.

In methodological terms, we conducted two experiments in which we manipulated a retail chatbot's orientation (selling orientation vs. customer orientation) in a service encounter. The second experiment also manipulated agent identity (chatbot vs. human) to examine if the non-human identity of a chatbot would produce different results compared to a human frontline service worker. The specific setting in both experiments was online retailing with respect to cosmetics. According to the McKinsey report *State of Beauty* (2025), online channels are expected to account for nearly one-third of global beauty sales (e.g., skin care products, cosmetics, and hair care products) by 2030 (up from 26 percent in 2024).

Taken together, the current study addresses the following main gap in relation to the existing literature: previous studies have focused on chatbots that provide service, and thereby they have neglected that agents representing the firm may engage in selling activities. The study is based on the notion of anthropomorphism (cf. Epley et al., 2007); it is expected that a non-human agent engaging in humanlike activities, such as making selling attempts, encourages the attribution of humanlike attributes to the agent. We hypothesize and demonstrate that attribution of warmth, competence, and theory of mind to the agent can explain how a selling-oriented approach and a customer-oriented approach produce different levels of customer satisfaction. These mediating variables have hitherto not been assessed in the literature on the effects of the two orientations, so the present study is intended to contribute to our understanding of *how* the orientations produce effects on outcome variables. More importantly, given that non-human agents need theory of mind to be truly useful for us humans (Cuzzolin et al., 2020; Langley et al., 2022; Matta, 2026), which is encouraging researchers to make attempts to create artificial theory of mind (e.g., Vinanzi et al., 2019), there is a general need to understand the factors that make us attribute theory of mind to such agents. We contribute to this understanding by showing how a chatbot's service encounter behavior – in terms of its selling orientation and customer orientation – can provide cues for attribution of theory of mind to the chatbot.

2. Conceptual framework and hypotheses

2.1. Overview of the framework

The starting point in our framework is the customer's exposure to a chatbot, which is applying one of two orientations (selling orientation vs. customer orientation) in the interaction with the customer. The chatbot's orientation is assumed to influence two social perception variables: warmth and competence. In the next step, it is assumed that each of these two variables influences the customer's attribution of theory of mind to the chatbot. With respect to the order of the mediators, we assume that warmth and competence are immediate person perception reactions (Fiske et al., 2007) and thus that they occur before attribution of theory of mind. Then, in the final step, attribution of theory of mind to the chatbot is assumed to impact customer satisfaction. The assumed links are depicted in Fig. 1.

2.2. Selling orientation and customer orientation

We conceptualize a chatbot engaging in selling activities in terms of a selling orientation which, in the existing literature, is typically seen in contrast to a customer orientation (e.g., Brown et al., 1991; Hansen et al., 2016; Locander et al., 2023; Thomas et al., 2001; Saxe and Weitz, 1982).

A selling orientation entails explicit selling attempts, such as cross selling and upselling, as well as putting pressure on the customer, painting a too rosy picture of products, and the use of compliance tactics (Brown et al., 1991; Locander et al., 2023; Thomas et al., 2001). It seeks to stimulate demand, it emphasizes the salesperson's or the firm's short-term goals (Hansen et al., 2016), and it is focused on closing the deal more than the development of long-term customer relationships (Kadic-Magljajlic et al., 2017).

In contrast to this, the core of a customer orientation is to help the customer make purchase decisions so that they satisfy the customer's needs in a long-term way (Kadic-Magljajlic et al., 2017; Locander et al., 2023; Saxe and Weitz, 1982). It comprises helping customers assess their needs, describing products adequately, avoiding high pressure selling, and avoiding the use of compliance tactics (Brown et al., 1991; Thomas et al., 2001). Moreover, a customer orientation can be seen as the practicing of the marketing concept at the level of the individual salesperson (Hansen et al., 2016; Kadic-Magljajlic et al., 2017).

In the literature, the two orientations are often seen as polar opposites (Kadic-Magljajlic et al., 2017), although it has been argued that they can and do co-exist (Guenzi et al., 2011). We employ the two orientations as the point of departure for our hypotheses; we view them as two separate dimensions (i.e., each can take on values from low to high) of a chatbot's behavior in an encounter with the customer.

2.3. The two orientations and the influence on warmth and competence

From the customer's point of view, and given that a non-human agent is somewhat humanlike, which can trigger anthropomorphism of the agent (Epley et al., 2007), we assume that a non-human agent's selling orientation and customer orientation provide cues about the agent in terms of two general social perception dimensions: warmth and competence.

Warmth has to do with the extent to which an agent's intentions are seen as good or bad, which can be indicated by the agent's friendliness, politeness, sincerity, and helpfulness (Fiske et al., 2007; Fiske, 2018). Competence is about the extent to which an agent is seen as able to enact his or her intentions; this can be indicated by the agent's intelligence, knowledge, expertise, confidence, assertiveness, and efficacy (Fiske et al., 2007; Fiske, 2018). The general assumption is that perceived warmth and competence of a target individual determine how we think about and treat this individual (Simon et al., 2020). Moreover, the two dimensions have been used in several studies when it comes to

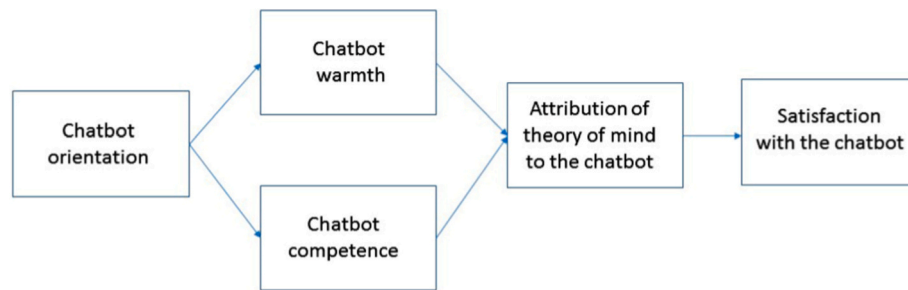


Fig. 1. Variables and links in the conceptual framework.

perceptions of chatbots (e.g., Lee and Li, 2023; Kim and Hur, 2024; Kull et al., 2021; Xu et al., 2023). The assumptions in the current study regarding the influence of an agent's orientation on the two social perception variables are as follows.

First, with respect to warmth, a selling orientation can signal that an agent is mainly interested in itself (Goad and Jaramillo, 2014), has a low concern for others (Itani et al., 2022), and has short-term personal gains as a main objective (Locander et al., 2023). This should be seen in contrast to a customer orientation which, by definition, emphasizes a concern for the customer's needs (Locander et al., 2023; Saxe and Weitz, 1982). In the light of this, we expect that the two orientations differ with respect to the extent that they signal an agent's self-focus versus other-focus. A high level of self-focus comprises an emphasis on the agent's own needs and feelings (Neff and Harter, 2003), and this type of agent is typically viewed as less empathic and less willing to help others (Lei et al., 2021). Given that general cues for warmth (e.g., friendliness, politeness, and helpfulness) are likely to reach lower levels for a self-focused agent, we expect the following outcome for the impact of the two orientations on warmth perceptions in the case of a chatbot:

H1a. When a chatbot uses a selling-oriented approach, it produces a lower level of perceived chatbot warmth compared to when it uses a customer-oriented approach

Second, when it comes to competence, our initial assumption is that there is a dominant customer-related dictum in many contemporary firms (and in the public sector), typically expressed in terms such as “the customer is king” and “the customer is always right”. This dictum, sometimes labelled customer authority (Keat et al., 2005) and the cult of the customer (du Gay and Salaman, 1992), can be seen as an overriding institutional imperative (du Gay and Salaman, 1992) and a main (Western) service philosophy (Kim and Aggarwal, 2016). That is to say, it is a customer orientation, not a sales orientation, which the dictum emphasizes. Indeed, salesperson objectives have changed from the time when Saxe and Weitz launched the notion of the two orientations, in the sense that a customer orientation has become the key element in firms' marketing activities (Homburg et al., 2011). We expect that the contemporary customer has been subject to a prolonged exposure to this dictum in the aggregate (as a customer and most likely as an employee, too) and can assess if an individual agent's service encounter behavior is aligned with the dictum. We also expect that the customer's view of an agent's alignment with the dictum provides competence cues in terms of the agent's perceived intelligence, knowledge, and expertise.

More specifically, we expect that (a) the selling-oriented agent is seen as misaligned with the dictum and is therefore perceived as having a lower level of competence and (b) the customer-oriented agent is seen as aligned with the dictum and is therefore perceived as having a higher level of competence. Hence the following is hypothesized for the impact of the two orientations on perceived competence in the case of a chatbot:

H1b. When a chatbot uses a selling-oriented approach, it produces a lower level of perceived chatbot competence compared to when it uses a customer-oriented approach

2.4. Warmth and competence and the impact on attributions of theory of mind

In the next step of the customer's attempts to make sense of a chatbot, we assume that perceptions of warmth and competence influence attribution of theory of mind to the chatbot. Theory of mind, sometimes referred to as mentalizing, is the ability to ascribe mental states to oneself and others, such as beliefs, perceptions, thoughts, emotions, desires, goals, and intentions (Baron-Cohen, 1999; Premack and Woodruff, 1978). This is indeed a central ability for many activities in social settings – including the employee's encounter with customers. More specifically, in the initial phase of the encounter, the employee needs to identify the customer's interests, goals and needs (Homburg et al., 2011). This, however, is not possible without the ability to ascribe mental states to the customer. Perhaps more importantly, having theory of mind is useful also for actively changing the future (Ho et al., 2022), so with theory of mind the employee can intervene in the life of the customer by offering him or her satisfaction of needs.

The customer, however, has no direct access to the employee's mind. Yet, the customer can attribute theory of mind to the employee based on various behavioral cues (e.g., what the employee says in a conversation with a customer). Indeed, even non-human agents are typically attributed some level of theory of mind if they display behaviors that are somewhat humanlike (Söderlund, 2022, 2024). We expect that perceptions of a chatbot's warmth and competence provide the customer with cues for attributions of theory of mind to the chatbot. More specifically, we assume that each of these perceptions is positively associated with attributions of theory of mind.

First, when a chatbot is displaying warmth, such as being polite and friendly, it indicates that the chatbot is making attempts to understand one specific mental state of the customer – the customer's needs. Second, when the chatbot displays competence in terms of acknowledging that it understands and has adopted the dominant customer-related dictum (“the customer is king”), it indicates that it can deliver an offer with the potential to satisfy the customer's needs. That is to say, both a display of warmth and a display of competence are likely to signal that the chatbot understands that another mind (i.e., the customer's mind) is present, and that this mind is in a state characterized by a gap between what exists and what is desired.

In other words, both warmth and competence perceptions provide cues that the chatbot's information processing is focused on the customer's mental states, which indicates that the chatbot has the capacity for ascribing mental states to the customer (i.e., the chatbot has theory of mind). From the customer's point of view, this is an aspect of perceived social attention; it is about the extent to which the customer perceives that he or she is in the focus of the chatbot's attention. In human-to-human contexts, we humans are typically very sensitive to others' attention; we can easily determine if we are attended to (Frischen et al., 2007; Reddy, 2003). Here, in the current study, it is assumed that others' attention to our own mind states is a primary marker of others' theory of mind.

In terms of previous studies, Kim and Hur (2024) show that both

chatbot warmth and chatbot competence are positively associated with perceived chatbot empathy, which they defined as the ability to sense, understand, and share the thoughts and feelings of another person. This empathy construct thus comprises the ability to attribute mental states to others (which is the core of having theory of mind).

Given H1_a and H1_b (i.e., a chatbot's selling orientation produces lower levels of perceived chatbot warmth and competence than the chatbot's customer orientation), and given the impact of warmth and competence on attributions of theory of mind just mentioned, the following is hypothesized:

H2. When a chatbot uses a selling-oriented approach, this produces a lower level of attributed theory of mind to the chatbot compared to when it uses a customer-oriented approach

2.5. The overall evaluation of the chatbot

The dependent variable in the current study is customer satisfaction with the chatbot, which can be seen as an overall post-interaction evaluation variable. Given many positive business-related consequences of customer satisfaction, such as increased customer loyalty, reduced price elasticities, insulation of current customers from competitive efforts, and lower costs of future transactions (Anderson et al., 1994), we view customer satisfaction as a relevant outcome variable for retailers.

We assume a positive impact of attribution of theory of mind to an agent on satisfaction with the agent. One main reason is that theory of mind is an essential human capacity (Baron-Cohen, 1999). Therefore, we expect that (a) attribution of theory of mind to a non-human agent contributes to making the agent appear more humanlike (Söderlund, 2022). Then, in the next step, we expect that (b) humanlikeness of a non-human agent is positively associated with the evaluation of the agent. That is to say, "what-is-humanlike-is-good" is a common response to non-human agents (Söderlund and Oikarinen, 2021). Empirically, Söderlund (2024) reports a positive association between attribution of theory of mind to a service robot and satisfaction with the robot. Given H2 (a chatbot's selling-oriented approach produces a lower level of attributed theory of mind to the chatbot compared to a customer-oriented approach), and given a positive impact of the attribution of theory of mind to a chatbot on customer satisfaction, the following is hypothesized:

H3. When a chatbot uses a selling-oriented approach, this produces a lower customer satisfaction level compared to when it uses a customer-oriented approach

2.6. Mediation assumptions

Our reasoning regarding H1-H3 implies that the impact of a chatbot's orientation on customer satisfaction can follow two routes (see Fig. 1). This means that the impact can be seen in mediation terms and as comprising a warmth-based route (i.e., orientation – warmth – attribution of theory of mind – satisfaction), and a competence-based route (i.e., orientation – competence – attribution of theory of mind – satisfaction). To assess this empirically, we hypothesize the following:

H4a. The impact of a chatbot's orientation on customer satisfaction is serially mediated by warmth and attribution of theory of mind to the chatbot

H4b. The impact of a chatbot's orientation on customer satisfaction is serially mediated by competence and attribution of theory of mind to the chatbot

2.7. Invariance assessment: agent identity

H1-H4 were tested in Study 1 in which the orientation of a retail chatbot was manipulated (selling orientation vs. customer orientation).

It should be noted that the hypothesized reactions are based on an anthropomorphism assumption. That is to say, given a somewhat humanlike chatbot, it is assumed that human reactions to the chatbot are similar to the reactions that a real human would evoke when it comes to warmth, competence and attribution of theory of mind.

To examine this empirically, H1-H4 were re-tested in Study 2, in which agent identity (chatbot vs. human employee) was a manipulated factor, too. Intuitively, agent identity may have a moderating influence on the hypothesized associations. For example, the specific orientation used by a human employee in a service encounter may provide stronger cues for the customer about the employee's warmth and competence (i.e., social perception variables originally developed for a human-to-human social setting) compared to a specific orientation's ability to provide cues about a non-human agent's warmth and competence. Yet, the potential for a human employee to come across as a full, distinct human in a chat context, in which the exchange between a sender and a receiver comprises only text, may be limited. Given this, specific hypotheses were not developed for the identity factor and its role was assessed in empirical terms in Study 2.

3. Study 1

3.1. Research approach and stimuli

We used a between-subjects experiment to test H1-H4 in a service encounter taking place on the website of a cosmetics retailer. This setting should be seen in the light of a growing market for products in the beauty category – and an increasing share of online sales for such products. It is expected that specialty retailers and mono-brand stores are particularly likely to maintain their market shares, according to the McKinsey report *State of Beauty* (2025).

More specifically, we asked the participants to imagine that they need new lipstick and that they visit the website of an online retailer of cosmetics. On that site, they find a specific lipstick (Yves Saint Laurent Loveshine) that they think is interesting, and while they examine information about it more closely, a window on the webpage pops up with this message: "Try Mary, our AI-Powered Digital Service Agent, if you need help with your shopping!". Then, they were told that they decided to test this agent and what followed was a scripted text-based conversation that we manipulated with respect to Mary's orientation (selling orientation vs. customer orientation). In the selling orientation condition, Mary engaged in both cross selling and upselling attempts, showed little interest in identifying the customer's specific needs, and used several traditional sales tactics. In the customer orientation condition, Mary made no explicit selling attempts and made an effort to understand the customer's specific needs (see the Supplementary Material for a full description).

Thus, we used a role-play scenario approach for the manipulations, which is common in research on service encounters – particularly when they involve non-human agents (e.g., Lu and Zhang, 2025; Söderlund, 2024; Xu et al., 2023; Zhang et al., 2025). As for the realism aspect, we used an existing product and we made the agent talk about it in terms of the same attributes that the manufacturer uses in its online product information.

3.2. Data collection and participants

The participants ($n = 282$; $M_{age} = 44.85$; only women were invited to the study, all were UK residents), who we recruited from Prolific, were randomly allocated to being exposed to one of the two orientations. The decision to include only women (and a female gendered chatbot) was based on the assumption that lipstick products are used primarily by women. In addition, we wanted to reduce the potential for confounds of opposite-sex dyadic behaviors in service encounters, such as flirting (Seeger-Guttmann and Medler-Liraz, 2023) and sexually-motivated behaviors (Wan and Wyer Jr, 2015). We collected the participants'

responses online with questionnaire items in Qualtrics. Our attention check item was formulated as follows: “Here is an attention task for you, just to make sure that you continue to read all questions carefully. Please select the number eight below”. The response alternatives were 5, 8, 12 and 15. No participant failed this and thus no participant was excluded from the analysis.

3.3. Measures

Scales with 10 points were used for the measures of all variables in the hypotheses, and Cronbach's alpha (CA), composite reliability (CR), and average variance extracted (AVE) were computed to assess measurement properties. We used SmartPLS 4.0 to assess discriminant validity with the HTMT approach (no HTMT values were $>.90$). An overview of the measurement items is provided in the Appendix.

Warmth was measured with the adjective pairs “cold–warm”, “unfriendly–friendly”, “impolite–polite”, and “unkind–kind” (CA = .96, CR = .96, AVE = .87). *Competence* was measured with the adjective pairs “incompetent–competent”, “unprofessional–professional”, “low knowledge–high knowledge”, “novice–expert”, and “unintelligent–intelligent” (CA = .93, CR = .95, AVE = .82). These items have been used in previous chatbot research by Söderlund and Mårtensson (2025) and Xu et al. (2023).

Moreover, we measured *attribution of theory of mind* to the chatbot (Mary) with the items “Mary is able to comprehend how others perceive things”, “Mary is able to understand what is in the mind of others”, “Mary is able to see things from others' point of view”, and “Mary is able to understand what humans want” (1 = do not agree at all, 10 = agree completely; CA = .96, CR = .96, AVE = .87). Similar items have been used in Söderlund (2024).

Our measure of *customer satisfaction* with the chatbot comprised three items adapted from Fornell (1992): “To what extent are you dissatisfied or satisfied with Mary?” (1 = very dissatisfied, 10 = very satisfied), “To what extent does Mary meet your expectations?” (1 = not at all, 10 = totally), and “Imagine a digital agent that is perfect in every respect. How near or far from this ideal do you find Mary?” (1 = very far from, 10 = cannot get any closer; CA = .97, CR = .97, AVE = .95). To assess the validity of the satisfaction measure, we used the item “Please write one word that captures your overall view of Mary” followed by an open response space. Then, we used the LIWC software to compute the emotional tone for each participant's selected word. Emotional tone can take on values in the 1–100 range; the higher the value, the more positive the emotional valence of the word (cf. Tausczik and Pennebaker, 2010). The resulting tone variable (M = 46.56, SD = 48.91) was positively and significantly correlated with the satisfaction variable ($r = .81$, $p < .01$). Given that customer satisfaction in a service encounter can influence intentions to buy, we included also an intention question related to the specific lipstick that was in focus in the stimulus material: “How likely is it that you would buy an Yves Saint Laurent Loveshine Lipstick from the online retailer represented by Mary?” (1 = very unlikely, 10 = very likely). The zero-order correlation between our customer satisfaction measure and the intention measure was positive and significant ($r = .85$, $p < .01$), which indicates validity in the satisfaction measure.

As a manipulation check, we used the 10-item version of the SOCO scale developed by Thomas et al. (2001). It has been used also by Kadic-Magljajic et al. (2017) and Locander et al. (2023). The scale comprises five items to measure selling orientation and five items to measure customer orientation. In our case, these items were used to capture a selling orientation: “Mary tries to sell as much as she can rather than satisfying customers”, “Mary stretches the truth in her sales presentations”, “Mary tries to sell as much as she can to convince the customer to buy”, “Mary paints a too rosy picture of the products to make them sound as good as possible”, and “Mary makes recommendations based on what she thinks she can sell and not on the basis of customers' long-term satisfaction” (1 = do not agree at all, 10 = agree completely, CA = .96). For customer orientation, we used the items

“Mary tries to figure out customers' needs”, “Mary has the customer's best interests in mind”, “Mary has a problem-solving approach in customer interactions”, “Mary recommends products that are best suited for customer needs”, and “Mary tries to find which products would be most helpful to customers” (1 = do not agree at all, 10 = agree completely, CA = .96). This, then, resulted in two separate orientation variables. It should be noted that they were negatively and significantly correlated in the sample ($r = -.75$, $p < .01$). A negative correlation between the two orientations is consonant with Anaza et al. (2023), Kadic-Magljajic et al. (2017), Locander et al. (2023), and the meta-analysis conducted by Goad and Jaramillo (2014).

3.4. Analysis and results

The manipulation check for the two orientations showed that the perceived selling orientation of the chatbot reached a higher level in the selling-orientation condition (M = 8.64, SD = 1.48) than in the customer-orientation condition (M = 3.89, SD = 1.86). This difference was significant (two-sided test, $t = 23.74$, $p < .01$, Cohen's $d = 2.82$). Moreover, the perceived level of the chatbot's customer orientation was higher in the customer-orientation condition (M = 8.55, SD = 1.27) than in the selling-orientation condition (M = 4.67, SD = 2.43). This difference was significant, too (two-sided test, $t = 16.81$, $p < .01$, Cohen's $d = 2.03$). Thus, the manipulation worked as intended.

With respect to H1_a, perceived warmth was lower in the selling-orientation condition (M = 5.79, SD = 2.15) than in the customer-orientation condition (M = 8.43, SD = 1.54). This difference was significant (two-sided test, $t = 11.85$, $p < .01$, Cohen's $d = 1.41$). Thus, H1_a was supported. Moreover, the level of perceived competence was lower in the selling-orientation condition (M = 6.61, SD = 2.06) than in the customer-orientation condition (M = 8.43, SD = 1.49). This difference was significant (two-sided test, $t = 8.53$, $p < .01$, Cohen's $d = 1.01$). This means that H1_b was supported.

As for H2, the level of attribution of theory of mind was lower in the selling-orientation condition (M = 3.71, SD = 2.27) than in the customer-orientation condition (M = 6.69, SD = 2.08). Since this difference was significant (two-sided test, $t = 11.50$, $p < .01$, Cohen's $d = 1.37$), H2 was supported.

Moreover, customer satisfaction was lower in the selling-orientation condition (M = 3.65, SD = 2.50) than in the customer-orientation condition (M = 7.70, SD = 1.69). This difference was significant (two-sided test, $t = 15.95$, $p < .01$, Cohen's $d = 1.90$), so H3 was supported.

The mediation hypotheses (H4_a and H4_b) were tested with the PROCESS macro (5000 bootstrap samples, Model 6). The experimental condition was the independent variable (scored as 1 = selling-orientation condition, 2 = customer-orientation condition). Customer satisfaction was the dependent variable. In the test of the warmth-based route of influence, warmth and attribution of theory of mind were serial mediators. This analysis showed that the indirect effect was significant (indirect effect: $b = .80$, SE = .14, CI 95% = [.54; 1.09]). The direct effect was significant, too (direct effect: $b = 1.33$, SE = .20, CI 95% = [.94; 1.72]). With respect to the competence-based route, competence and attribution of theory of mind were serial mediators. Also in this case, the indirect effect was significant (indirect effect: $b = .65$, SE = .11, CI 95% = [.45; .87]). The direct effect was significant (direct effect: $b = 1.69$, SE = .19, CI 95% = [1.32; 2.06]). Both routes of mediation, then, were of the complementary type (cf. Zhao et al., 2010) and H4_a and H4_b were supported.

To assess the robustness of the H4 tests, we used the system of variables depicted in Fig. 1 as the proposed model in a SEM-based analysis with SmartPLS 4.0. To this proposed model we added three (non-hypothesized) links (orientation–satisfaction, warmth–satisfaction, and competence–satisfaction) to be able to assess mediation along the lines of what Sarstedt et al. (2020) suggest for a mediation analysis. In this model, R^2 for satisfaction was .82. With this approach, the warmth-based route of influence remained significant. That is to say, there was

a significant indirect effect of the orientation variable on customer satisfaction via warmth and attribution of theory of mind ($b = .18$, $t = 3.68$, $p < .01$). Again, the direct effect of the orientation variable on satisfaction was significant ($b = .46$, $t = 5.93$, $p < .01$), thus suggesting complementary mediation. This provides support for H4a. The competence route of influence remained significant, too. The indirect effect of the orientation variable on satisfaction, via competence and attribution of theory of mind, was significant ($b = .17$, $t = 4.04$, $p < .01$). Since the direct effect was also significant, this indicates mediation of the complementary type for the competence-based route of influence, and it provides support for H4b.

3.5. Discussion

The main result in Study 1 was that customer satisfaction was attenuated when the chatbot used a selling orientation as opposed to a customer orientation. The results also show that a mechanism for this influence involved perceived warmth, competence, and attribution of theory of mind to the chatbot.

With respect to the relation between warmth and competence, it can be noted that both of them reached high levels in the overall sample. In a human-to-human setting, this “high-high” combination has been obtained for target groups such as the middle class (Fiske, 2018). Nevertheless, the overall sample level of perceived chatbot competence was higher ($M = 7.53$, $SD = 2.01$) than the overall level of perceived chatbot warmth ($M = 7.12$, $SD = 2.29$). This difference was significant (two-sided test, $t = 4.46$, $p < .01$). Higher scores for competence than for warmth have been obtained for human salespersons in the assessment made by Strinić et al. (2022), so in that sense our results reflect that anthropomorphism appears to have taken place among the participants.

In an attempt to examine if the results would be replicated, we conducted a second study with another specific product in focus and with other participants. In addition, as indicated above, the second study involved the manipulation of agent identity (chatbot vs. human employee).

4. Study 2

4.1. Research approach and stimuli

Again, as in Study 1, we used a between-subjects experiment to test the hypotheses in a service encounter taking place on the website of a cosmetics retailer. The product in focus in Study 2, however, was a specific perfume (Yves Saint Laurent Libre Eau de Parfum). The orientation of the agent, called Mary also in Study 2, was manipulated (selling orientation vs. customer orientation) in the same way as in Study 1. Agent identity (chatbot vs. human employee) was an additional manipulated factor. In the chatbot conditions, Mary was presented as an “AI-Powered Digital Service Agent”; in the human employee conditions, Mary was introduced as one of the retailer's most experienced beauty advisors, and she presented herself as spending half of her time with customers in stores and the other half online. Moreover, she stated that “I have been working with cosmetics for about a decade, and I have tried many of the products that we sell” (see the Supplementary Material for full descriptions). Thus, we employed a role-play scenario approach also for Study 2, and it comprised an existing product that the agent talked about in terms of the same attributes that the manufacturer uses in its own online product information.

4.2. Data collection and participants

The participants ($n = 422$; $M_{age} = 43.54$; again, only women were invited to the study; all were UK residents), recruited from Prolific, were randomly allocated to being exposed to one of the four conditions. We collected the participants' responses with questionnaire items in Qualtrics. The same attention check as in Study 1 was used; no participant

failed to answer correctly and thus no participant was excluded from the analysis.

4.3. Measures

Warmth ($CA = .95$, $CR = .96$, $AVE = .88$), competence ($CA = .94$, $CR = .96$, $AVE = .81$), attribution of theory of mind ($CA = .97$, $CR = .97$, $AVE = .91$), and customer satisfaction ($CA = .98$, $CR = .98$, $AVE = .96$) were measured with the same items as in Study 1. Discriminant validity was assessed with the HTMT approach and SmartPLS 4.0 (no HTMT values were $>.90$). An overview of the items is provided in the Appendix.

To assess the validity of the satisfaction measure, we used the same approach with emotional tone and purchase intention measures as in Study 1. The emotional tone variable ($M = 41.38$, $SD = 48.34$) was positively and significantly correlated with the satisfaction variable ($r = .86$, $p < .01$). Moreover, the correlation between the satisfaction variable and the intention variable (“How likely is it that you would buy an Yves Saint Laurent Libre Eau de Parfum from the online retailer represented by Mary?”; 1 = very unlikely, 10 = very likely) was positive and significant ($r = .88$, $p < .01$). Taken together, then, this indicates validity in our customer satisfaction measure.

For the manipulation check of the agent's orientation, we used the same items as in Study 1 for a selling orientation ($CA = .95$) and a customer orientation ($CA = .96$). In Study 2, the zero-order correlation between the two orientation variables was $-.74$ ($p < .01$).

Moreover, for the manipulation check of the identity factor, we assumed that the capacity to make free, autonomous choices is a distinct human capacity (cf. Feldman et al., 2016). In other words, given that freedom is experienced as a consequence of attributing action to a self (Kruglanski and Cohen, 1973), an entity with a self would be attributed a higher level of freedom than an entity without a self. And a human is expected to be seen as having more of a self than a non-human. Therefore, we expected that an observer (here: the participant) would attribute a higher level of free will to a human compared to a non-human. We used items developed by Nadelhoffer et al. (2014) to assess people's beliefs in free will as a point of departure, and we adapted them to a situation in which the task of the participants was to assess Mary's level of free will: “Mary had complete control of what she said in the conversation”, “What Mary said was based on her own choices about what to say”, “Mary was in charge of what she should say”, “Mary decided herself what to say” and “Mary had free will when it comes to what to say in the conversation” (1 = do not agree at all, 10 = agree completely, $CA = .94$).

4.4. Analysis and results

As a check of the orientation manipulation, we used a 2 X 2 ANOVA with the two manipulated factors as independent variables. When perceived selling orientation was the dependent variable, there was a main effect of the orientation factor ($F = 613.71$, $p < .01$, $\eta^2 = .59$). The level of perceived selling orientation was higher in the selling-orientation conditions ($M = 8.44$, $SD = 1.35$) than in the customer-orientation conditions ($M = 4.20$, $SD = 2.09$). Neither the identity factor nor the interaction were significant. Moreover, there was a significant main effect of the orientation factor when perceived customer orientation was the dependent variable ($F = 411.91$, $p < .01$, $\eta^2 = .50$). In this analysis, the level of the perceived customer orientation was higher in the customer-orientation conditions ($M = 8.47$, $SD = 1.40$) than in the selling-orientation conditions ($M = 4.69$, $SD = 2.31$). The identity factor and the interaction were not significant. In addition, we made a qualitative manipulation check by examining the most frequent words used by the participants for their own description of the agent. In the selling orientation conditions, the most frequent word was “pushy”; in the customer orientation conditions, the most frequent word was “helpful”. The manipulation of the orientation factor, then, worked as

intended.

As for the check of the agent identity manipulation, we used a 2 X 2 ANOVA with the manipulated factors as independent variables and perceived agent free will as the dependent variable. This resulted in a significant main effect of the identity factor ($F = 46.58$, $p < .01$, $\eta^2 = .10$). The orientation factor and the interaction were not significant. As expected, the perceived level of agent free will was lower in the chatbot conditions ($M = 5.02$, $SD = 2.50$) than in the human employee conditions ($M = 6.59$, $SD = 2.19$). Thus, the identity manipulation worked as intended.

H1-H3 were tested with the same 2 X 2 ANOVA approach as above. For warmth, there was a main effect of the orientation factor ($F = 176.20$, $p < .01$, $\eta^2 = .30$). The identity factor and the interaction were not significant. The level of warmth was lower in the selling-orientation conditions ($M = 6.02$, $SD = 2.25$) than in the customer-orientation conditions ($M = 8.49$, $SD = 1.49$). This means that H1_a was supported. Similarly, for competence, the orientation factor was significant ($F = 84.41$, $p < .01$, $\eta^2 = .17$). The identity factor and the interaction were not significant. Competence reached a lower level in the selling-orientation conditions ($M = 7.09$, $SD = 1.90$) than in the customer-orientation conditions ($M = 8.66$, $SD = 1.56$). This means that H1_b was supported.

When it comes to attribution of theory of mind, there was a main effect of the orientation factor ($F = 182.24$, $p < .01$, $\eta^2 = .30$). Neither the identity factor nor the interaction were significant. The level of attributed theory of mind was lower in the selling-orientation conditions ($M = 3.73$, $SD = 2.21$) than in the customer-orientation conditions ($M = 6.64$, $SD = 2.22$). This means that H2 was supported. Although the identity factor was not significant, the human employee was attributed a higher level of theory of mind ($M = 5.39$, $SD = 2.59$) than the chatbot ($M = 4.97$, $SD = 2.70$), and we will return to this result in the general discussion section.

For customer satisfaction, the main effect of the orientation factor was significant ($F = 357.01$, $p < .01$, $\eta^2 = .46$). The identity factor and the interaction were not significant. The level of satisfaction was lower in the selling-orientation conditions ($M = 3.70$, $SD = 2.39$) than in the customer-orientation conditions ($M = 7.71$, $SD = 1.94$). Thus, H3 was supported.

In the mediation analysis of H4_a and H4_b, we used the PROCESS macro (5000 bootstrap samples, Model 6) with the orientation factor as the independent variable (scored as 1 = selling-orientation condition, 2 = customer-orientation condition) and customer satisfaction as the dependent variable. To test the warmth-based route of influence, warmth and attribution of theory of mind were serial mediators. The analysis showed that the indirect effect was significant (indirect effect: $b = .80$, $SE = .11$, $CI\ 95\% = [.60; 1.02]$). The direct effect was significant, too (direct effect: $b = 1.42$, $SE = .16$, $CI\ 95\% = [1.11; 1.73]$). In the analysis of the competence-based route, competence and attribution of theory of mind were serial mediators. The indirect effect was significant (indirect effect: $b = .56$, $SE = .09$, $CI\ 95\% = [.40; .74]$). The direct effect was significant, too (direct effect: $b = 1.75$, $SE = .15$, $CI\ 95\% = [1.44; 2.05]$). Thus, both routes of mediation were of the complementary type (cf. Zhao et al., 2010), and H4_a and H4_b were supported.

The robustness of the H4 tests was assessed with SmartPLS 4.0 in the same way as in Study 1 (R^2 for satisfaction was .82). In this analysis, there was a significant indirect effect of the orientation variable on customer satisfaction via warmth and attribution of theory of mind ($b = .15$, $t = 4.56$, $p < .01$). The direct effect of the orientation factor on satisfaction was significant ($b = .50$, $t = 8.52$, $p < .01$), thus suggesting complementary mediation. This provides support for H4_a. There was also a significant indirect effect of the orientation factor on satisfaction via competence and attribution of theory of mind ($b = .10$, $t = 3.75$, $p < .01$). Since the direct orientation-satisfaction effect was significant, this indicates mediation of the complementary type for the competence-based route of influence, and it provides support for H4_b.

4.5. Discussion

Study 2 replicated the Study 1 results with another product (a perfume), with other participants, and with an additional manipulated factor. The main result was that the agent's use of a selling-oriented approach in the service encounter attenuated customer satisfaction compared to a customer-oriented approach. Again, as in Study 1, the impact of the orientation on customer satisfaction was mediated by warmth, competence, and attribution of theory of mind. Agent identity did not moderate the effects of the agent's orientation, and identity per se did not produce a main effect on the variables in the hypotheses.

With respect to the relation between warmth and competence for the chatbot, Study 2 resulted in the same "high-high" combination as in Study 1. Yet, as in Study 1, the overall sample level of perceived chatbot competence was higher ($M = 7.84$, $SD = 1.94$) than the overall level of perceived chatbot warmth ($M = 7.31$, $SD = 2.25$). This difference was significant (two-sided test, $t = 5.61$, $p < .01$). Thus, again, the participants' responses to the chatbot mirror responses to human salespersons as captured in a study by Strinić et al. (2022).

5. General discussion

5.1. The selling-oriented versus customer-oriented chatbot and its effects

Overall, our results show that a chatbot's orientation in the service encounter affects the level of satisfaction with the chatbot. The results also show that this effect can be explained in terms of warmth, competence, and attribution of theory of mind as mediation variables. The effect size of the impact of orientation type was generally large for each outcome variable, in terms of conventional standards, and all effect sizes were larger than the average explained variance (i.e., .06) in consumer research (Eisend, 2015).

With respect to the selling orientation and its negative impact on customer satisfaction, our result is consonant with findings showing that a selling orientation can have this effect in a human-to-human context (e.g., Goff et al., 1997). The result showing a positive effect of a customer orientation on satisfaction is consonant with a positive effect of this orientation on the customer's evaluation of the employee (Homburg et al., 2011), satisfaction (Brady and Cronin Jr, 2001; Goff et al., 1997), value creation (Guenzi et al., 2011), and repurchase intentions (Kadic-Magljajlic et al., 2017) in human-to-human contexts.

In contrast to existing studies, however, the current study comprised not only a non-human agent but also (confirmed) hypotheses about mediation with variables hitherto not examined in the literature on the two orientations. For example, our results are consonant with the findings in Homburg et al. (2011) showing that the (human) employee's customer orientation boosts customers' evaluations of the employee, but in their case, there were no empirical data about mediators to explain this effect. As for conceptual contributions, then, the current study offers evidence of *how* a chatbot's orientation can affect the evaluation of the chatbot.

Warmth and competence were originally launched as two general social perception dimensions in a human-to-human context (Fiske et al., 2007). They have also been applied to brands (e.g., Kervyn et al., 2022; Liu et al., 2022) and various forms of non-human agents (e.g., Kim and Hur, 2024; Kull et al., 2021; Söderlund and Mårtensson, 2025). In relation to this research, the current study contributes by showing that the orientation of an agent (here: the way an agent conducts a text-based conversation) has the capacity to influence perceptions of the agent's warmth and competence. It may be noted that it has been argued that warmth is judged before competence, that warmth judgments carry more weight in affective and behavioral reactions, and that warmth is more predictive than competence of future benefits of a relationship (Eisenbruch and Krasnow, 2022; Fiske et al., 2007). One main reason is that the primacy of warmth is fitting from an evolutionary perspective; another person's intent for good or ill is more important to survival than

whether the person can act on those intentions (Fiske et al., 2007). It has also been argued that warmth-related traits of a target tend to be more accessible and predictive. Therefore, they are more heavily weighted in evaluations of the target (Kull et al., 2021). The primacy of warmth is corroborated in the current study, in the sense that the indirect impact of the orientation factor on customer satisfaction was stronger for the warmth-based route than for the competence-based route (i.e., $b = .80$ vs. $b = .65$ in Study 1; $b = .80$ vs. $b = .56$ in Study 2).

One aspect of the warmth-competence relation is worth noting: as reported above, both studies resulted in a “high-high” combination. For human targets, this combination is typically associated with pride and admiration for the targets (Fiske, 2018). Nevertheless, in both studies, the overall sample level of perceived chatbot competence was significantly higher than the overall level of perceived chatbot warmth. This, then, indicates that the participants in the current study were less generous in attributing the capacity to have intentions for good or ill than in attributing an ability to act on those intentions to a non-human agent. Given that behavior (here: what the chatbot said in the conversation) is more indicative of agency compared to a behavioral intention, the participants’ responses reflect that we humans are sensitive to agency-related signals (Barrett and Johnson, 2003).

Moreover, the current study indicates that the orientations influence attribution of theory of mind via warmth and competence. The following should be noted before this is commented on more in detail: several existing studies have examined attribution of mind to non-humans (i.e., they have examined the extent to which non-humans are perceived to have mental states). Typically, this is done in terms of what has been proposed as two main dimensions of mind, namely agency and experience (Gray et al., 2007). With this view, agency has to do with the capacity for self-control, morality, memory, planning and thought; experience is about the capacity to feel hunger, fear, pain and pleasure (ibid.). Generally, studies of attribution of mind to a target along these two dimensions show the same pattern as in Gray et al.’s (2007) seminal study, namely that (1) humans are attributed more mind than non-humans, and (2) non-human agents, such as robots, are attributed more agency than experience. The current study, however, has been concerned with attribution of theory of mind, not attribution of mind, and the former is a more sophisticated capacity. That is to say, if (a) I think that you want one particular product, I have attributed a mental state to you (I think that you have wants) and thus I have attributed mind to you. In contrast to this, if (b) I believe that you think that I want a particular product, I have assumed that you are capable of attributing mental states to me and thus I have attributed theory of mind to you (cf. Söderlund, 2024). Yet, to date, relatively few studies have examined attribution of theory of mind to non-humans (ibid.). This represents a gap in relation to the existing literature, given that many observers believe that non-human agents need theory of mind in order to interact effectively with humans (Matta, 2026; Williams et al., 2022). An additional gap, however, is that the literature on artificial theory of mind typically neglects perceptual aspects. That is to say, several authors have discussed various architectures for developing theory of mind in non-humans (e.g., Williams et al., 2022; Vinanzi et al., 2019), presumably with the assumption that there is a direct link between an agent having theory of mind and humans’ perceptions that this is the case. However, in a service encounter, this link may not be direct: the agent needs to display its capabilities and the customer needs to recognize and interpret what the agent displays. The net result, in the customer’s mind, may not be the same as what the developer of artificial theory of mind has equipped the agent with. The current study is an attempt to contribute to this perceptual aspect by showing that a chatbot’s service encounter behaviors can influence customers’ attribution of theory of mind to the chatbot.

Study 2 provides an opportunity to compare the level of theory of mind that the participants attributed to the chatbot and to the human employee. As noted in the result section for Study 2, the impact of the identity factor on attribution of theory of mind was not significant. Yet,

the human employee was attributed a higher level of theory of mind ($M = 5.39$) than the chatbot ($M = 4.97$). This, then, indicates a belief that a human employee has a better understanding of humans’ mental states than a chatbot. What is worth noting is that the chatbot was attributed a non-trivial level of theory of mind, which indicates that anthropomorphism did take place. It should also be noted that the human employee was not attributed a full theory of mind (given the 10-point scale that we used to measure attribution of theory of mind). This is the same result as in Söderlund (2024), and it can be seen as a form of dehumanizing the employee. There are at least three possible reasons for this. First, we do not have full access to someone’s ability to ascribe mental states to others. Second, a service/sales job in retailing is not a high-status job; the employee can be seen as a means to an end for the customer and thereby as relatively mindless (Epley, 2018). Third, in the current study, the employee appeared in a chat window and was only visible as a sender and receiver of text on a screen. This may have provided the participants with relatively limited cues about the employee’s humanness.

Moreover, Study 2 comprised an additional mind-related variable, namely attribution of free will to the agent (we used it as a manipulation check for the identity factor). Free will is closely related to agency – thus, the attribution of free will to an agent is likely to be related to the attribution of agency to an agent. As mentioned above, previous research shows that humans are attributed more agency than non-humans (Gray et al., 2007) and that non-humans are attributed non-trivial levels of agency if they display humanlike behaviors (e.g., van der Woerd and Haselager, 2019). Indeed, although most people would not believe that non-humans, such as chatbots and service robots, have an actual will of their own, an intuitive inclination to attribute agency to a non-human may be hard to suppress (van der Woerd and Haselager, 2019). This is reflected in our Study 2 chatbot conditions in the sense that the perceived free will of the chatbot was non-negligible ($M = 5.02$ on a 10-point scale). This, then, is another indication that anthropomorphism did take place in relation to the chatbot.

Finally, the current study indicates that attribution of theory of mind to the agent had a positive influence on customer satisfaction with the agent. In general, an ability to attribute mental states to others is an essential human capability, which makes it possible to explain and predict (and manipulate) others’ behavior (Epley et al., 2007; Ho et al., 2022). The ability to attribute mental states to others is essential in a service encounter: when the agent has this ability, it can understand that a customer has specific needs. And to satisfy the customer’s needs is a fundamental principle in a customer-oriented approach (Itani et al., 2022; Saxe and Weitz, 1982). Seen in this light, a high level of customer satisfaction would be difficult to achieve in the absence of at least some ability for an agent to understand the customer’s mental states. Moreover, an additional aspect may contribute to the influence of attribution of theory of mind to an agent on satisfaction with the agent: the customer’s belief that an agent considers his or her mental states can signal a sense of importance (or “mattering”). In the meaningfulness-in-life literature, this has to do with the extent to which a person believes that his or her existence is valuable (George and Park, 2017). Importance is a valenced state of mind – the feeling that one is important is positively associated with positive affect and life satisfaction (ibid.). Conversely, not having one’s mental states considered (i.e., they are ignored) is a highly negatively charged state of mind. Indeed, few things in life are worse than the feeling that one is irrelevant to others (Elliot et al., 2004). In any event, the current study contributes to the customer satisfaction literature by demonstrating that attribution of theory of mind to an agent is an additional antecedent of satisfaction in service encounters.

5.2. Practical implications

In a service encounter setting, the specific agent with whom the customer interacts is the company from the customer’s point of view

(Bitner et al., 1990). That is to say, the impression of the agent is typically carried over to the company in a valence-congruent way. In the light of this, our results imply that a selling-oriented approach should be avoided and that a customer-oriented approach should be encouraged.

One way to accomplish this in practice is to use theory specifying the main ingredients in the two orientations (cf. section 2.2) as the point of departure for instructions to chatbots. If the chatbot is powered by generative AI (GenAI) and has no-code programming options, employees can train the chatbot through a conversation (cf. Wirtz and Stock-Homburg, 2025) in which it is exposed to the conceptual core of the two orientations. Moreover, with GenAI-enabled no-code programming, a non-human agent can learn from real-time interactions and immediately update its algorithms based on feedback (ibid.). In the case of feedback on chatbots' usage of the two orientations, post-interaction surveys to customers can include the SOCO scale items developed by Thomas et al. (2001) that we used in the current study. Such surveys can be used to monitor the behavior of chatbots; the survey responses can be seen as facilitating behavior-based sales controls that have been used for human employees (cf. Zheng et al., 2023). In addition, caution is called for in the training of chatbots. If they are trained in an unsupervised way, with real employee-customer interaction data, one may assume that such interactions comprise elements of a selling-oriented approach (e.g., cross selling and upselling). This aspect could be dealt with by feedback mechanisms comprising conceptual reference points so that chatbots' suboptimal behaviors are mitigated.

A customer-oriented approach entails the identification of the customer's interests, goals, and needs (Homburg et al., 2011), which can be personal and sensitive, particularly in the context of cosmetics, so there is a need for caution when it comes to consumer privacy. That is to say, in the process of identifying customers' needs, the chatbot is likely to collect data that may be subject to privacy concerns. Such concerns involve the customer's worries about the loss of control of information about himself or herself and about what might happen with this information in a future situation (Solove, 2006). Since privacy concerns can produce negative effects on the overall evaluations of a non-human agent (Söderlund, 2023), there is a need to handle customer data generated by a customer-oriented approach in a privacy-preserving way and to explicitly communicate this in the interaction with the customer.

Moreover, given that warmth judgments carry more weight in affective and behavioral reactions than competence judgments (Fiske et al., 2007), which is what the current study indicates (i.e., in both experiments, the indirect impact of the orientation factor on customer satisfaction was stronger for the warmth-based route than for the competence-based route), it may be possible to actively steer chatbots in a warmth direction by their use of language. One main indicator of warmth is politeness (Fiske et al., 2007), which means that explicit politeness strategies can be helpful. Showing interest, avoiding disagreement, and asserting common ground in a conversation are some examples (Brown and Levinson, 1987).

It should also be underlined that a customer orientation can produce several negative consequences for the individual (human) service employee. They include requirements to internalize feelings that are experienced as false, the adoption of a submissive mindset, and exposure to new forms of managerial control and surveillance, which can result in stifled spontaneity, stress, emotional numbness, alienation from self, and burnout (du Gay and Salaman, 1992; Sturdy, 1998). These effects, however, will not occur for non-human agents engaging in customer-oriented activities in the service encounter.

Yet, our results raise questions regarding the future of selling. Given that human service employees feel uncomfortable with being salespersons (de Ruyter et al., 2020), and given the results in the current study (i.e., customers react with reduced satisfaction when a non-human agent makes sales attempts), one wonders who will be the salesperson of the future.

5.3. External validity aspects

The current study explored effects of a chatbot's behavior on customer satisfaction by manipulating the chatbot's selling orientation versus a customer orientation in a setting comprising service encounters online, with respect to cosmetics, and female consumers. From an external validity point of view, however, the specifics of this setting call for elaboration.

One aspect is the extent to which the results would generalize to a similar setting outside the study. As we see it, two reasons make this likely. First, the results with respect to customer satisfaction (i.e., a selling orientation attenuates customer satisfaction and a customer orientation boosts customer satisfaction) are consonant with prior results from a human-to-human setting (Brady and Cronin Jr, 2001; Goff et al., 1997; Homburg et al., 2011). Second, as noted above, the effect sizes produced by the orientation manipulations were generally large, so it could perhaps be argued that we have somehow "overcharged" the manipulations. We do not think so; we think that we have stayed true to the conceptual content of both orientations. More importantly, and in human-to-human service encounters, many frontline employee behaviors in the service encounter are causally potent. Indeed, research since several decades has produced a long list of employee behaviors that influence customer satisfaction (e.g., Bitner et al., 1990; Frazer Winsted, 2000). Some of these behaviors are relatively less salient than those displayed in a selling orientation and in a customer orientation; nevertheless, they do influence customer satisfaction. This indicates that we humans are sensitive to what other humans do to us, even when it comes to minor activities (and it is presumably this sensitivity that makes it possible for a somewhat humanlike non-human agent to produce similar effects). In other words, given that both orientations represent a bundle of behaviors, it is not so surprising that the combined behaviors in the bundle can generate strong effects.

Another aspect has to do with the extent to which the results would generalize to settings with other products than cosmetics, which have several specific characteristics: they are self-expressive, hedonically motivated, and are means to improve the user's attractiveness (Bloch and Richins, 1992). Yet, some other products have similar characteristics. Holbrook et al. (1986) have assessed customers' views of the purpose served by a range of products, including lipstick (our Study 1) and perfume (our Study 2), and they identified these two products as being high when it comes to aesthetic and symbolic purposes. Other products with a similar position in Holbrook et al. (1986) are shoes, dresses, swimsuits, hats, sunglasses, liquor, and cheese. Clothes and accessories are frequently bought online, and in that sense our results may generalize to service encounters involving products from this category.

Moreover, the current study comprised only women, because cosmetics are a ubiquitous element of women's consumer culture and represent a fundamental way in which women invent and transform themselves (Coulter et al., 2003). The same is not the case for men. This focus on female participants, however, means that we do not know if men would react in similar ways when they are exposed to a selling orientation and a customer orientation. It should also be recalled that the stimulus chatbot was female gendered, which calls for additional aspects to keep in mind in further studies. More specifically, opposite-sex dyads may introduce distinct customer reactions in a service encounter (e.g., perceptions of flirting and sexually-motivated behaviors; Seger-Guttmann and Medler-Liraz, 2023; Wan and Wyrer Jr, 2015). Such reactions, depending on their valenced charge for the customer, may boost or attenuate the effects of an agent's orientation on customer satisfaction.

5.4. Other limitations and suggestions for further research

Our manipulations of the two orientations were based on the idea that each orientation in the experiment should have the same conceptual content as in theory on the orientations. Since each orientation

comprises several agent behaviors, however, this means that the manipulations may be conceived of as expressions of high levels of both orientations. Yet, at least conceptually, each orientation can be seen as ranging from low to high. Further research, then, can allow both orientations to have low versus high levels. It is possible to use also other types of orientations in the experimental conditions as a contrast to a selling orientation. For example, a selling orientation can be contrasted with a service orientation, and a chatbot can be designed to emphasize educational and/or entertainment aspects in its conversations with users (Sands et al., 2021). In addition, given problems in many firms with employees who are uncomfortable with engaging in both selling and providing service (de Ruyter et al., 2020), it would be valuable to examine an explicit ambidexterity condition (i.e., a hybrid form of selling orientation and customer orientation) as a control condition. Further studies, then, can benefit from including additional ways to conceptualize a chatbot's orientation.

Another issue related to chatbot behavior that merits attention is that some authors view a sales attempt in the service encounter as service and not as a sales activity. For example, according to Polonsky et al. (2000), sales attempts (i.e., suggesting the purchase of a related product) may contribute to perceptions of better service levels. Such attempts may be compatible with a customer orientation; the customer may interpret them as an agent's genuine understanding of what the customer actually needs. However, other authors have found that sales attempts (i.e., upselling) result in adverse effects on customers' future brand patronage (Norvell et al., 2018). Similar results have been reported for cross selling and its effects on customer satisfaction (Ku and Huang, 2016). To come to terms with this, it would be useful to conduct experimental studies in which sales attempts are manipulated in such a way that the agent suggests versus does not suggest that the customer should buy a particular product discussed in a service encounter.

Moreover, whatever a chatbot is doing in terms of sales activities and other activities, it can be seen as scripted. Previous empirical studies in a human-to-human context suggest that this can influence the customer. It has been shown, for example, that the perceived "scriptedness" of a frontline service employee is negatively associated with customers' perceptions of service quality (Nguyen et al., 2014). In the current study, our measure of perceived agent free will (in Study 2) can be seen as an indirect indication of the extent to which the agent followed a conversation script (i.e., a high level of free will is assumed to indicate a low level of "scriptedness"). And our results show that there was no significant difference in perceived free will between the selling-orientation conditions and the customer-orientation conditions. A more direct measure of "scriptedness", however, may be able to pick up differences between orientations. Indeed, studies in which such measures are used could reveal if participants view a customer orientation as scripted – and this is a possible outcome, given the prevalence of this type of orientation.

It also needs to be observed that the stimulus interaction between the agent and the customer was made to mirror a typical online chat conversation, which means that it involved an exchange of text between invisible parties. Chatbots, however, are beginning to appear with a (humanlike) face and a voice. This opens the door for various non-verbal communication cues that can influence the customer in the service encounter (Gabbott and Hogg, 2000). Presumably, such cues can interact with a chatbot's orientation when it comes to effects on the customer. It is possible, for example, that a chatbot with an appealing face and a soft, gentle voice is perceived in less negative terms when it uses a selling-oriented approach. Further research, then, can explore the effects of various orientations when non-verbal cues are taken into account.

As for the identity factor (i.e., human vs. chatbot), which did not influence the response variables (Study 2), it should be acknowledged that our results were obtained for a service encounter taking place in a pre-purchase phase. That is to say, we instructed the participants to assume the role of a customer engaged in online browsing. In the event

of a purchase, however, there is a possibility that the identity factor can become more influential. More specifically, a non-human agent may not understand what a purchase means in the same way as a human agent. What a purchase means, among other things, is that it can create pain of payment for the buyer. This has to do with the negative affect that consumers experience when they become cognizant that they have lost some of their financial resources, or when they become aware that they will or may lose some of their financial resources in the future. Pain derived in this way can play a significant role in consumer decision-making (Reshadi and Fitzgerald, 2023). A purchase means also transfer of ownership. Ownership is probably an intuitively clear concept for most humans, although its abstract functions as a core mechanism for controlling, using, and disposing of assets may be hard to articulate for lay persons. In any event, given that a non-human agent cannot feel pain, and cannot own anything, its activities in service encounters involving a purchase may produce different effects than those produced by a human agent. Indeed, examinations of non-human agents' perceived understanding of purchases and ownership, which represent fundamental human phenomena, would advance our knowledge of human reactions to non-humans.

Turning now to boundary conditions, in terms of individual differences that may moderate the impact of orientation type on customer satisfaction, it should first be noted that many products (including cosmetics; cf. Guthrie and Kim, 2009) can be subject to different levels of user involvement. The high personal relevance characterizing high involvement is expected to motivate a higher level of information seeking and more extensive needs to compare and evaluate products (Zaichkowsky, 1985). Given that a selling orientation entails a stronger focus on the seller's needs than on the customer's needs, it is possible that high involvement customers view a selling orientation as less informative. In other words, given high customer involvement, the negative effects of a selling orientation on customer satisfaction may be boosted in relation to low involvement. Therefore, further research is needed to examine if the effects obtained in the current study would be the same under different levels of customer involvement.

Moreover, it is possible to view a selling orientation as an expression of agent dominance and a customer orientation as an expression of agent submission. Acknowledging that customers can vary in terms of their general need for dominance and submission in service encounters, and according to Ma and Dubé (2011), a service encounter can result in two main combinations: complementary dyadic behaviors (a dominant agent and a submissive customer; a submissive agent and a dominant customer) and anti-complementary dyadic behaviors (a dominant agent and a dominant customer; a submissive agent and a submissive customer). And these combinations have implications for customer satisfaction. For example, a submissive agent and a submissive customer can produce a state of mutual passiveness, which impedes the customer's task-oriented goals and results in a negative impact on customer satisfaction (Ma and Dubé, 2011). In contrast, a dominant agent and a submissive customer may create a more efficient dyad when it comes to executing service tasks, and this can boost customer satisfaction (ibid.). To explore these possibilities when it comes to the effects of an agent's orientation on customer satisfaction, researchers can introduce an explicit submission-dominance variable to capture both the customer's perceptions of the agent and what level the customer desires for him-/herself in a service encounter.

An additional individual difference variable that the current study ignored has to do with a sense of importance. In the theory section, we argued that the customer's beliefs that an agent considers his or her mental states can signal a sense of importance (or "mattering") for the customer. We also argued that a sense of importance can contribute to a positive impact of beliefs about the agent's theory of mind on customer satisfaction. Yet, a sense of importance can be seen as a stable trait, which is subject to individual variation, in the same way as personality (Fino et al., 2024). Given this, one may assume that those who are high on this trait are less sensitive to an agent's orientation than those who are

low. That is to say, if one's general sense of importance is already high, a momentary reduction in a sense of importance produced by an agent's selling orientation in a service encounter may not matter much. In contrast, if one's sense of importance is low, a selling-oriented agent may serve to further stress one's insignificance – particularly if the agent is non-human (“I do not even matter to non-humans”). Thus, a sense of importance, as a general trait, may moderate the impact of an agent's orientation and customer satisfaction.

Finally, we used a purchase intention variable as a validity check for our customer satisfaction variable (and these two variables were positively and significantly correlated). Further studies of the effects of a chatbot's selling-related activities, however, should be designed so that actual purchases can be captured. Previous research shows, for example, that cross selling attempts can be highly influential (e.g., Milligan and Hantula, 2006). They indeed represent cost-efficient means for an employee to create sales in a service encounter. Therefore, the potential for a selling orientation to generate more immediate sales than a customer orientation needs to be examined before the final word can be said about the effects of the two orientations.

CRediT authorship contribution statement

Magnus Söderlund: Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Formal analysis, Conceptualization. **Meriem Moutawakil Andersson:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Measurement items in Study 1 and Study 2

Warmth

What is your impression of Mary, the digital service agent?
Cold–warm
Unfriendly–friendly
Impolite–polite
Unkind–kind.

Competence

What is your impression of Mary, the digital service agent?
Incompetent–competent
Unprofessional–professional
Low knowledge–high knowledge
Novice–expert
Unintelligent–intelligent.

Attribution of theory of mind

Mary is able to comprehend how others perceive things.
Mary is able to understand what is in the mind of others.
Mary is able to see things from others' point of view.
Mary is able to understand what humans want.

Customer satisfaction

To what extent are you dissatisfied or satisfied with Mary?
To what extent does Mary meet your expectations?
Imagine a digital agent that is perfect in every respect. How near or far from this ideal do you find Mary?

Selling orientation

Mary tries to sell as much as she can rather than satisfying customers.
Mary stretches the truth in her sales presentations.
Mary tries to sell as much as she can to convince the customer to buy.
Mary paints a too rosy picture of the products to make them sound as good as possible.
Mary makes recommendations based on what she thinks she can sell and not on the basis of customers' long-term satisfaction.

Customer orientation

Mary tries to figure out customers' needs.
Mary has the customer's best interests in mind.
Mary has a problem-solving approach in customer interactions.
Mary recommends products that are best suited for customer needs.
Mary tries to find which products would be most helpful to customers.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jretconser.2026.104805>.

Data availability

Data will be made available on request.

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